



100+ AI INTERVIEW QUESTIONS



VRAM Math



FlashAttn



MoE & GQA



vLLM Serving

- ✓ From GPU memory calculation to Speculative Decoding.
- ✓ Master vLLM, GQA, MoE & Production LLM Serving .

VRAM & KV CACHE
MEMORY ENGINE

FLASH-ATTN 3
SRAM IO SPEED

DPO VS PPO
RLHF ALIGNMENT

Swipe to ace your AI Engineering Interview



GPU VRAM Math: How to Serve a 70B Model on 80GB H100?

Model Weights (140GB
FP16)

→ Quant →

INT4 Weights (35GB)

+

KV Cache (24GB)

Model Memory Equation:

- **FP16 Weight Size:** $70 \times 2 \times \text{bytes} = 140 \text{ GB VRAM}$
(Requires 2x 80GB GPUs).
- **INT4 Quantization:** $70 \times 0.5 \times \text{bytes} = 35 \text{ GB VRAM}$ (Fits on single 80GB H100!).

KV Cache Allocation Formula:

- $\text{KV Cache Size} = 2 \times n_{\text{layers}} \times n_{\text{heads}} \times d_{\text{head}} \times \text{seq_len} \times \text{batch_size} \times \text{bytes}$.
- For Llama-3-70B with 4k context & batch 16, KV Cache takes **~24GB VRAM**.

 **Real-World:** Meta & vLLM use INT4 AWQ quantization to serve Llama-3-70B on 1x H100 GPU.

 **Fact:** Always reserve 20% VRAM buffer for activation memory spikes during peak load.

FlashAttention-3: How Block-Tiling Eliminates Memory IO

⚠ Standard Attention

Writes $N \times N$ attention matrix to HBM.
Memory IO bottleneck $O(N^2)$.

⚡ FlashAttention-3

Tiles Q, K, V blocks directly in fast GPU SRAM.
Zero HBM materialization.

Architecture & Math Mechanics:

- **HBM Memory Bottleneck:** GPU compute (TFLOPS) is 100× faster than memory transfer (HBM bandwidth).
- **Tiling + Online Softmax:** Recomputes attention scores in 256KB SRAM blocks without writing full attention matrix to HBM.
- **Performance Impact:** 3×–5× faster attention computation; scales long contexts up to 128k.



Real-World:

OpenAI GPT-4o & Anthropic Claude 3.5 use FlashAttention kernels for sub-50ms TTFT.

Speculative Decoding: 2.5x Faster Inference via Small Draft Models



Autoregressive generation is token-by-token memory-bound. Speculative decoding turns memory bound into compute bound.

Execution Workflow:

- 1. **Draft Step:** Fast 8B model generates $K=5$ candidate tokens sequentially ($<5\text{ms}$).
- 2. **Verification Step:** Large 70B Target model verifies all 5 tokens in a SINGLE parallel forward pass.
- 3. **Acceptance:** Accept verified tokens matching target probability distribution.

 **Fact:** Mathematically guarantees EXACT same output distribution as 70B model while boosting throughput by 2.5x.

Position Embeddings: RoPE vs ALiBi for 128k Context Scaling

Standard Absolute Positional Embeddings fail when input context exceeds training length. RoPE and ALiBi solve extrapolation.

1. Rotary Position Embeddings (RoPE):

- Encodes position by rotating query and key vectors in 2D complex plane: $R_{\{\Theta, m\}}^d x_m$.
- **RoPE Scaling (YaRN / Linear):** Interpolates rotation frequencies to stretch context from 4k $\rightarrow$ 128k tokens.

2. ALiBi (Attention with Linear Biases):

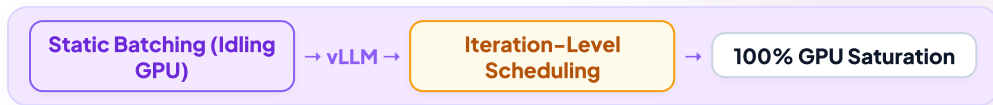
- Injects static negative bias matrix directly into attention scores based on distance: $-m \cdot |i - j|$.
- Zero position embeddings required; extrapolates infinitely without fine-tuning.



**Real-
World:**

Llama-3 & Mistral-Large use RoPE with YaRN frequency scaling for 128k context support.

Continuous Batching: How vLLM Eliminates GPU Bubble Time



Static batching waits for the longest sequence in a batch to finish, leaving GPU tensor cores idle (GPU Bubble).

Iteration-Level Scheduling (vLLM / TGI):

- **Dynamic Insertion:** Eject completed sequences immediately after token generation step.
- **Dynamic Injection:** Inject new incoming user requests into the active iteration loop.
- **PagedAttention:** Virtual memory paging for KV Cache allocates non-contiguous RAM pages, reducing VRAM fragmentation to <4%.

 **Real-World:** vLLM PagedAttention boosts production serving throughput by 4x over naive HuggingFace pipelines.

Quantization Trade-offs: **AWQ vs GPTQ vs GGUF**

AWQ (Activation-aware)

Protects top 1% salient weight channels. Zero loss in perplexity.

Naive RTN (Round-to-Nearest)

Rounds all weights blindly. Destroys model accuracy at 4-bit.

Production Quantization Formats:

- **AWQ (Activation-aware Weight Quantization):** Best for GPU vLLM serving. Preserves critical feature activations.
- **GPTQ (One-shot Layer-wise Quantization):** Fast 4-bit GPU inference using second-order Taylor expansion.
- **GGUF / llama.cpp:** CPU & Apple Silicon Metal quantized execution with hybrid CPU/GPU offloading.

 **Fact:** INT4 AWQ reduces VRAM footprint by 75% with less than 0.5% degradation in benchmark accuracy.

Mixture of Experts (MoE): Top-2 Routing & Capacity Factor

Input Token



Gating Router



Expert 2 & Expert 5

MoE scales model capacity ($8 \times 7B = 47B$ total params) while activating only 13B active params per token.

Gating & Routing Mechanics:

- **Top-2 Softmax Gating:** Router calculates probability $G(x) = \text{Softmax}(\text{TopK}(x \cdot W_g, 2))$ to select 2 experts.
- **Auxiliary Load Balancing Loss:** Penalizes router if tokens over-concentrate on Expert 1, preventing expert collapse.
- **Capacity Factor:** Hard buffer limit on max tokens per expert to prevent GPU thread stalling.

 **Real-World:** Mixtral-8x7B & DeepSeek-V3 use MoE to achieve GPT-4 performance at 1/4th the inference cost.

Attention Architecture: MHA vs MQA vs Grouped Query Attention (GQA)

8 Q Heads

→ Group →

1 Shared KV Pair

→

8× KV VRAM Savings

Comparison Matrix:

- **Multi-Head Attention (MHA):** \$\$\$ Query heads, \$\$\$ Key heads, \$\$\$ Value heads. Huge KV Cache memory footprint.
- **Multi-Query Attention (MQA):** \$\$\$ Query heads, 1 Key head, 1 Value head. 8× KV memory savings, slight accuracy drop.
- **Grouped Query Attention (GQA):** \$\$\$ Query heads grouped into \$\$\$ Key/Value pairs (e.g. 8 Query heads per 1 KV head).

⚡ **Impact:** GQA reduces KV Cache memory by 8× while matching MHA quality, enabling 8× larger batch sizes.

RLHF Alignment: DPO (Direct Preference Optimization) vs PPO

🔧 PPO (Classic RLHF)

Requires 4 separate neural nets (Actor, Critic, Reward, Ref). Unstable RL tuning.

⚡ DPO (Direct Alignment)

Eliminates Reward model! Optimizes policy directly on preference pairs (y_w, y_l) .

DPO Loss Function Formula:

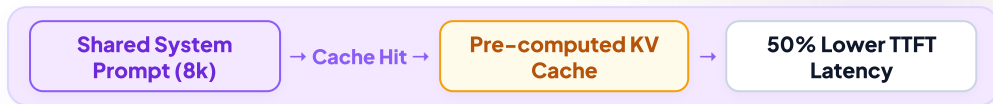
- $$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$
- Replaces complex actor-critic RL loop with a simple binary cross-entropy classification loss.



Real-World:

Llama-3 & Zephyr-7B use DPO alignment for stable, 2x faster preference training.

Prompt Caching: Reusing Prefix KV Cache Across Requests



Multi-turn agent chats repeat long system instructions on every turn. Prompt caching avoids re-computation.

Prefix Cache Mechanics:

- **Radix Tree Matching:** Store pre-computed KV tensors in a Radix Tree indexed by prompt token prefix hashes.
- **TTFT Speedup:** Time-To-First-Token (TTFT) drops from 800ms to <100ms on long prompt prefix matches.
- **Cost Discount:** Anthropic & OpenAI offer 50% API price discounts on cached prompt tokens.

 **Fact:** Prompt Caching saves up to 50% in token API costs for multi-turn agent workflows.

Structured Output: Grammar-Guided Constrained Sampling

Relying on prompt formatting ("Return valid JSON") fails randomly. Grammar-guided sampling guarantees 100% schema validity.

Constrained Sampling Mechanics (Outlines / Instructor):

- **CFG Grammar Masking:** Converts JSON Schema / Pydantic model into a Context-Free Grammar (CFG) Finite State Automaton.
- **Logit Masking:** Before sampling step, sets logit value = $-\infty$ for any token violating grammar rules.
- **Guaranteed Format:** Output is guaranteed to parse cleanly into Pydantic models with 0 retry loops.

Real-World: OpenAI Structured Outputs & Outlines framework enforce strict JSON schemas at logit sampling level.

Long Context vs RAG: 1M Window vs Vector Search Trade-offs

\$ 1M Context Window

Passes 1M tokens raw into prompt. High latency (2s+ TTFT) & \$5.00/query.

⚡ RAG + Reranking

Retrieves top 5 pristine chunks (2k tokens). Sub-200ms TTFT & \$0.01/query.

Needle-In-A-Haystack (NIAH) Limitation:

- Even 1M context models show recall degradation when multiple competing facts are present.
- **Engineering Recommendation:** Use RAG for large document retrieval; use Long Context for analyzing a single 50-page complex PDF file.

Distributed Parallelism: Tensor Parallelism (TP) vs Pipeline Parallelism (PP)

Tensor Parallel (Intra-Node NVLink)



Pipeline Parallel (Inter-Node InfiniBand)

Parallelism Strategies:

- **Tensor Parallelism (TP):** Splits matrix multiplication W across GPUs inside a single server node via NVLink (900 GB/s).
- **Pipeline Parallelism (PP):** Splits model layers across multi-node servers via InfiniBand network.
- **Sequence Parallelism (SP):** Splits activation memory across sequence length dimension.



**Rule of
Thumb:**

Use TP=8 within 8-GPU NVLink nodes; use PP=4 across separate multi-node server racks.

LLM Evaluation: LLM-as-a-Judge vs ELO Rating Benchmarks

Model A vs Model B



GPT-4o Judge



LMSYS Chatbot Arena
ELO

Traditional BLEU / ROUGE metrics fail to measure qualitative LLM generation accuracy.

Production Evaluation Frameworks:

- **LLM-as-a-Judge (GPT-4o):** Evaluates pairwise preference with position swapping to eliminate position bias.
- **LMSYS Chatbot Arena ELO:** Crowdsourced human preference voting calculating ELO scores ($\$R_A = R_A + K(S_A - E_A)\$$).

 **Real-World:** Top AI labs benchmark models using LMSYS ELO & MT-Bench suite.

Multi-Agent Systems: Supervisor Router vs ReAct Agent Loops

Supervisor Agent

→ Route →

Coder Agent

+

Tester Agent

Agent Architectural Patterns:

- **ReAct Pattern (Reason + Act):** Interleaves thought chains: `Thought` $\rightarrow$ `Action` $\rightarrow$ `Observation` $\rightarrow$ `Response`.
- **Supervisor Router Pattern:** Central manager agent evaluates task status and routes sub-tasks to specialized worker agents.
- **DAG Execution:** LangGraph state machines prevent agent infinite loops via directed acyclic graphs.



Real-
World:

LangGraph & AutoGen use Supervisor state machines for enterprise multi-agent workflows.

ACE YOUR AI INTERVIEW **LIKE A PRO**

From GPU VRAM math to vLLM, FlashAttention & Multi-Agent systems.

WHAT WE COVERED:

- VRAM Math, KV Cache Allocation & Quantization (AWQ/GPTQ)
- FlashAttention-3, Speculative Decoding & Continuous Batching
- RoPE/ALiBi, DPO Alignment & Multi-Agent Architecture

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